FINANCIAL MARKET ANALYSIS

Week 1 Project Plan: Data Acquisition & Preliminary Analysis

Python | Pandas | NumPy | Matplotlib

This document defines the proposed data sources, analysis questions, workflow, tools, quality checks, timeline, and expected outputs. It is a plan for the analysis; no findings or results are claimed at this stage.

1. Project Overview

The project will examine historical daily stock-market data for a small group of publicly listed companies. The purpose is to practice a complete early-stage analytics workflow: locating suitable data, retrieving it reproducibly, checking data quality, preparing useful variables, and conducting preliminary exploratory analysis. The initial scope will use three large, actively traded Indian-listed companies from different business sectors, subject to ticker and data availability checks. A common observation period of approximately three years will be used where possible. The final company list and date range will be recorded in the notebook after data availability is confirmed.

This is an educational descriptive-analysis project, not investment advice. Historical price patterns do not guarantee future performance, and the project will not make buy/sell recommendations or claim to predict future prices.

2. Objectives and Analysis Questions

- Acquire historical daily price and trading-volume data from public financial-data sources.
- Assess data completeness, consistency, duplicate records, missing values, and unusual values.
- Calculate daily percentage returns and summarize the distribution of returns for each company.
- Compare historical price movement and trading activity over the selected period.
- Estimate and compare rolling volatility as a descriptive indicator of how widely returns fluctuate.
- Create clear charts and a concise summary of observations, limitations, and possible next steps.

Key questions

- How did each selected stock's adjusted closing price change over the observation period?
- What were the average daily return, median daily return, and overall return variability for each stock?
- Which stock showed the greatest historical variation in daily returns during the chosen period?
- Were there visible periods of unusually large daily movements or changes in trading volume?
- How did rolling volatility change over time, and were the patterns similar across the selected stocks?

3. Public Data Sources

Source A — Yahoo Finance

Website: https://finance.yahoo.com/ | Python access: https://pypi.org/project/yfinance/

Yahoo Finance provides historical market information for many listed securities, including daily open, high, low, close, adjusted close where available, and volume. The Python package yfinance can be used to retrieve historical data into a Pandas DataFrame, which is convenient for a reproducible notebook-based workflow. The primary dataset will be obtained from this source if the chosen Indian stock symbols and date ranges are available.

Practical note: yfinance is a third-party Python interface rather than an official Yahoo Finance API. Data availability, field definitions, and access can change. The project will record the retrieval date, ticker symbols, requested period, and package version, and will review the source's applicable terms before use.

Source B — Alpha Vantage

Documentation: https://www.alphavantage.co/documentation/

Alpha Vantage provides documented financial-data endpoints, including daily time-series information with open, high, low, close, and volume fields. It is a secondary source for checking whether the selected instruments and data fields can be obtained independently, or for a small sample cross-check if its available access tier supports the required symbols and history.

Practical note: access requires an API key, and request limits and historical-data entitlements depend on the plan and endpoint. The project will not assume that every endpoint or the full observation period is available free of charge. If the secondary source is constrained, it will be documented as a limitation rather than delaying the core work.

Data selection criteria

- Choose three publicly listed Indian companies from different sectors, with consistent symbols and sufficient daily observations.
- Use the same start and end dates for comparisons wherever data availability permits.
- Prefer adjusted closing prices for return calculations where reliably provided, and document the selected field.
- Retain date, ticker, open, high, low, close, adjusted close (if available), and volume fields.
- Save a raw, unmodified copy of the retrieved data separately from the cleaned analysis dataset.

4. Data Acquisition Plan

- Confirm the three stock symbols and verify that each source returns data for the intended period.
- Set a fixed observation window of approximately three years, ending at the latest date consistently available across the chosen securities.
- Retrieve daily records from Yahoo Finance using yfinance; handle each ticker separately if necessary.
- Inspect the returned columns, date index, data types, row counts, and earliest/latest dates.
- Export the original data to CSV files in a raw_data folder. Record source, ticker, retrieval date, requested period, and any retrieval errors.
- Where feasible, request a small overlapping sample from Alpha Vantage and compare dates and comparable price fields. Explain any differences instead of assuming that two sources must match exactly.
- Keep API keys out of notebooks, source code, screenshots, and GitHub commits. Use environment variables or a local configuration file excluded from version control if an API key is used.

5. Data Cleaning and Preparation Plan

- Standardize column names and ensure dates are parsed as dates; sort records in ascending date order.
- Check for duplicate dates within each ticker and investigate duplicates before removing any records.
- Inspect missing values by column. Distinguish expected non-trading days from missing observations on dates when data should exist.
- Convert price and volume fields to numeric types and flag parsing errors or impossible values for review.
- Check that prices are positive and that high/low values are logically consistent with open/close values. Investigate anomalies rather than automatically deleting them.
- Check for large gaps in dates and compare a sample against the source where possible.
- Avoid filling missing market prices with arbitrary values. Exclude records from a calculation only when justified, and document the decision.
- Create a clean analysis dataset while preserving the raw files unchanged.

6. Preliminary Exploratory Analysis

The initial analysis will focus on descriptive statistics and visual inspection rather than formal forecasting. Calculations will be performed separately for each ticker and then compared over the common date range.

- **Summary statistics:** Count, mean, median, standard deviation, minimum, maximum, and quartiles for available price, volume, and return fields.
- **Daily returns:** Calculate daily percentage returns from the selected adjusted closing-price series where available: (current adjusted close / previous adjusted close − 1) × 100.
- **Price trends:** Plot each stock's adjusted closing price over time. A rebased index (first valid value = 100) may be used for easier comparison of relative movement across different price levels.
- **Return distribution:** Use histograms and summary statistics to inspect the spread and asymmetry of daily returns.
- **Rolling volatility:** Calculate a 20-trading-day rolling standard deviation of daily returns. Label this as a historical descriptive measure, not a forecast.
- **Trading volume:** Plot volume over time and examine whether unusually high-volume days coincide with notable price movements, without assuming causation.
- **Cross-stock comparison:** Compare cumulative historical return and return variability over the same observation window, clearly noting the period and metric used.

Planned visuals include a price-trend line chart, rebased price comparison, daily-return histograms, rolling-volatility chart, and trading-volume chart. Each chart will include a title, axis labels, units where appropriate, legend, and date range.

7. Python Libraries and Tools

Library / Tool	Planned use	Why it is suitable
Python	Main programming language	Readable, widely used for data analysis, and supports a reproducible workflow.
yfinance	Retrieve Yahoo Finance historical data	Provides a convenient interface for downloading historical market data into Pandas.
Pandas	Load, clean, join, filter, and summarize tabular data	DataFrames and date-aware operations make time-series data manageable.
NumPy	Numerical calculations and array operations	Supports efficient numeric work and complements Pandas calculations.
Matplotlib	Create time-series plots, histograms, and comparison charts	Produces customizable charts suitable for analysis and reporting.
Seaborn (optional)	Improve statistical plots if useful	Can make distributions and relationships easier to inspect; not essential to the first version.
Requests / JSON or CSV tools (optional)	Call a documented API or read downloaded files	Useful if Alpha Vantage is used for a source check; use only as needed.
Jupyter Notebook or Google Colab	Document code, checks, charts, and observations	Combines executable code with explanations and outputs.
Git and GitHub	Version control and portfolio presentation	Makes the project reproducible and allows code, documentation, and sample outputs to be shared.

8. Timeline and Milestones (Estimated 33 Hours)

Phase	Estimated effort	Milestone / output
Scope and source research	3 hours	Confirm objectives, source access, ticker symbols, date range, and field definitions.
Data acquisition	5 hours	Retrieve data, inspect responses, save raw CSVs, and record retrieval details.
Initial data audit	4 hours	Review schema, row counts, date coverage, data types, missing values, and duplicates.
Cleaning and preparation	6 hours	Standardize dates and fields, validate numeric values, resolve or document anomalies, and create clean data.
Preliminary analysis	7 hours	Calculate returns, descriptive statistics, rolling volatility, and cross-stock comparison measures.
Visualization and interpretation	4 hours	Prepare charts and write concise, evidence-based observations.
Validation, documentation, and GitHub preparation	4 hours	Re-run the workflow, check outputs, document limitations, prepare README, and organize files.

Total estimated effort: 33 hours. The schedule is effort-based rather than tied to calendar dates. If a source has access limits, the core analysis will proceed with the primary source and the limitation will be recorded.

9. Quality Checks, Risks, and Limitations

- **Source access or API limits:** Use Yahoo Finance as the primary source and Alpha Vantage only for feasible cross-checking; record limitations.
- **Different source values:** Check date alignment, adjusted versus unadjusted prices, currency, and field definitions before comparing.
- **Missing or duplicate records:** Profile the data, investigate the cause, and document any exclusions or transformations.
- **Market holidays and non-trading days:** Do not treat expected non-trading days as missing daily observations.
- **Corporate actions:** Prefer adjusted closing prices for return analysis where available and explain that data-provider adjustment methods may differ.
- **Overinterpretation:** Describe associations and historical patterns only; do not infer causation or present findings as financial advice.

10. Expected Deliverables

- A project-plan document defining the data sources, objectives, workflow, tools, timeline, and limitations.
- Raw CSV files with source and retrieval metadata.
- A cleaned dataset for the selected companies and a documented cleaning process.
- A Jupyter Notebook (.ipynb) containing acquisition steps, data-quality checks, calculations, visualizations, and interpretations.
- A README.md describing the project objective, setup, data sources, how to run the notebook, and key limitations.
- A small set of clearly labelled charts and a concise findings summary after the actual analysis has been completed.

11. Conclusion

This plan establishes a manageable and reproducible path from public financial data to preliminary insights. The first priority is to verify data availability and quality before drawing conclusions. Pandas will support data preparation, NumPy will support numerical operations, and Matplotlib will communicate patterns visually. The 33-hour estimate reserves time not only for coding but also for validation, documentation, and communicating limitations. The project can then be extended in later stages if appropriate, for example by adding a dashboard or more detailed sector-level comparisons.

12. References

- Yahoo Finance. Financial market data and historical data pages. https://finance.yahoo.com/
- yfinance project documentation and package information. https://pypi.org/project/yfinance/
- Alpha Vantage API Documentation. https://www.alphavantage.co/documentation/
- Pandas documentation. https://pandas.pydata.org/docs/
- NumPy documentation. https://numpy.org/doc/
- Matplotlib documentation. https://matplotlib.org/stable/